

Building an Influenza Forecasting Pipeline

SISMID Lab Overview

July 22 through July 24, 2026

A Human-AI Teaming Approach to Infectious Disease Modeling

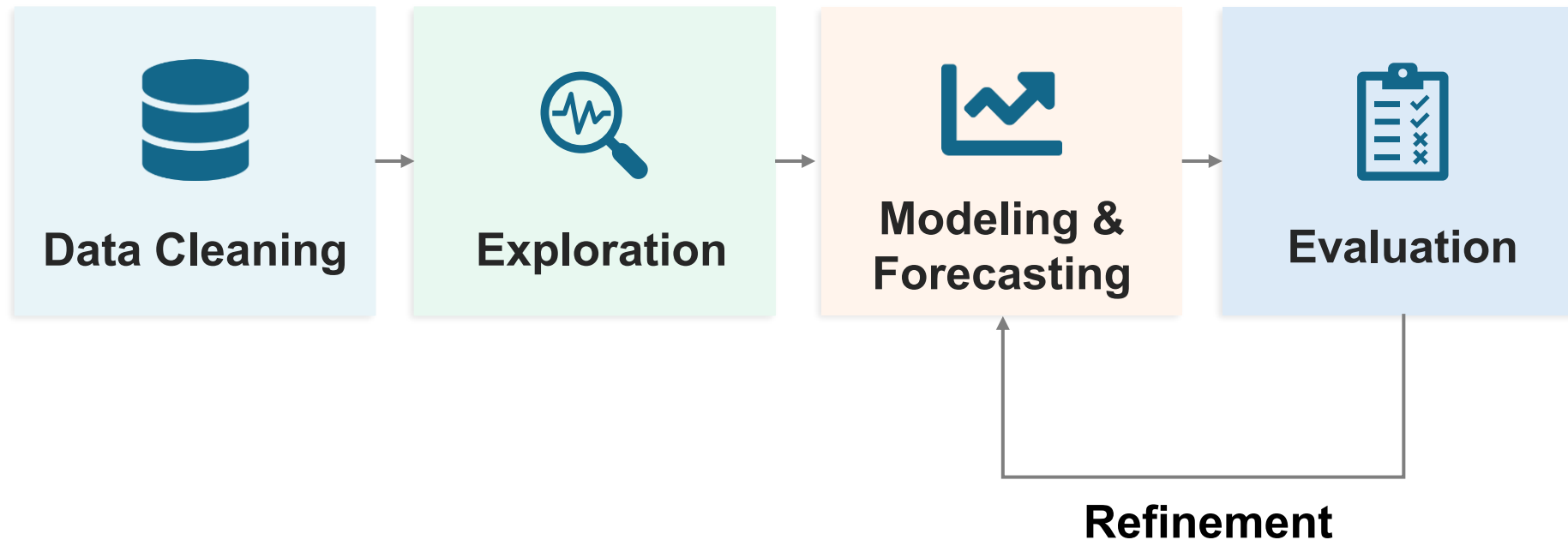
Overview

- Every week during influenza season, public health decision-makers need to anticipate what's coming:
 - How many people will be hospitalized next week, and in the weeks ahead?
 - When will the peak hit and how high will it be?
- In this lab, you'll work as a **human-AI team** to build a national-level influenza hospitalization forecasting system using real CDC data to help answer these questions.

You define the **vision** and **rules**. AI helps **build** and **implement** the solution.

What You're Building

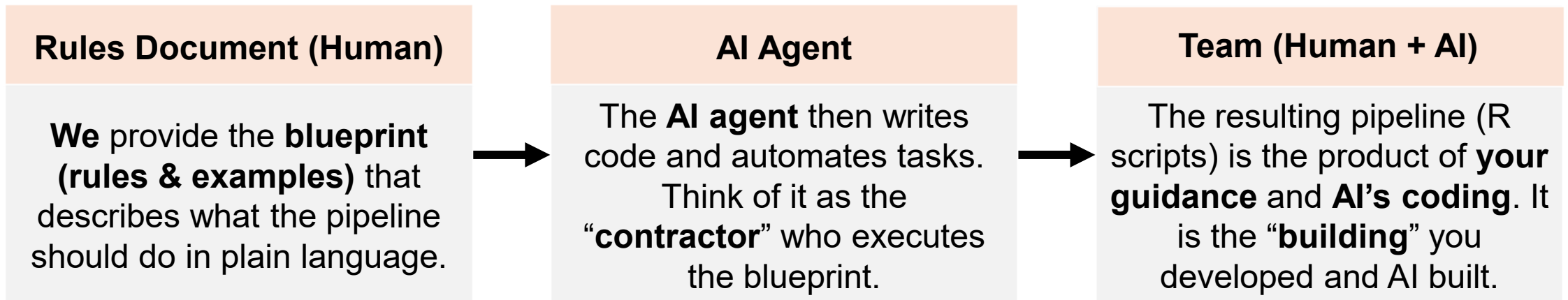
A **rules** document that evolves into a complete forecasting pipeline and a **FluSight-ready** model.



Rules Document: The **blueprint** that describes what the pipeline should do in **plain language**.

Where does AI come in?

- Think of AI like an **efficient contractor**: fast and helpful, but it still requires direction.
- It needs **rules** and **examples** to produce accurate output (the human component).
 - The clearer the guidance, the better the results.



Rules Document vs AI Agent

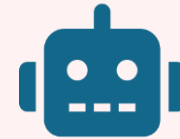


`rules.md`

Written by YOU

- Human-readable specifications
- Describes WHAT the pipeline should do
- You edit this to improve results

Copilot
generates



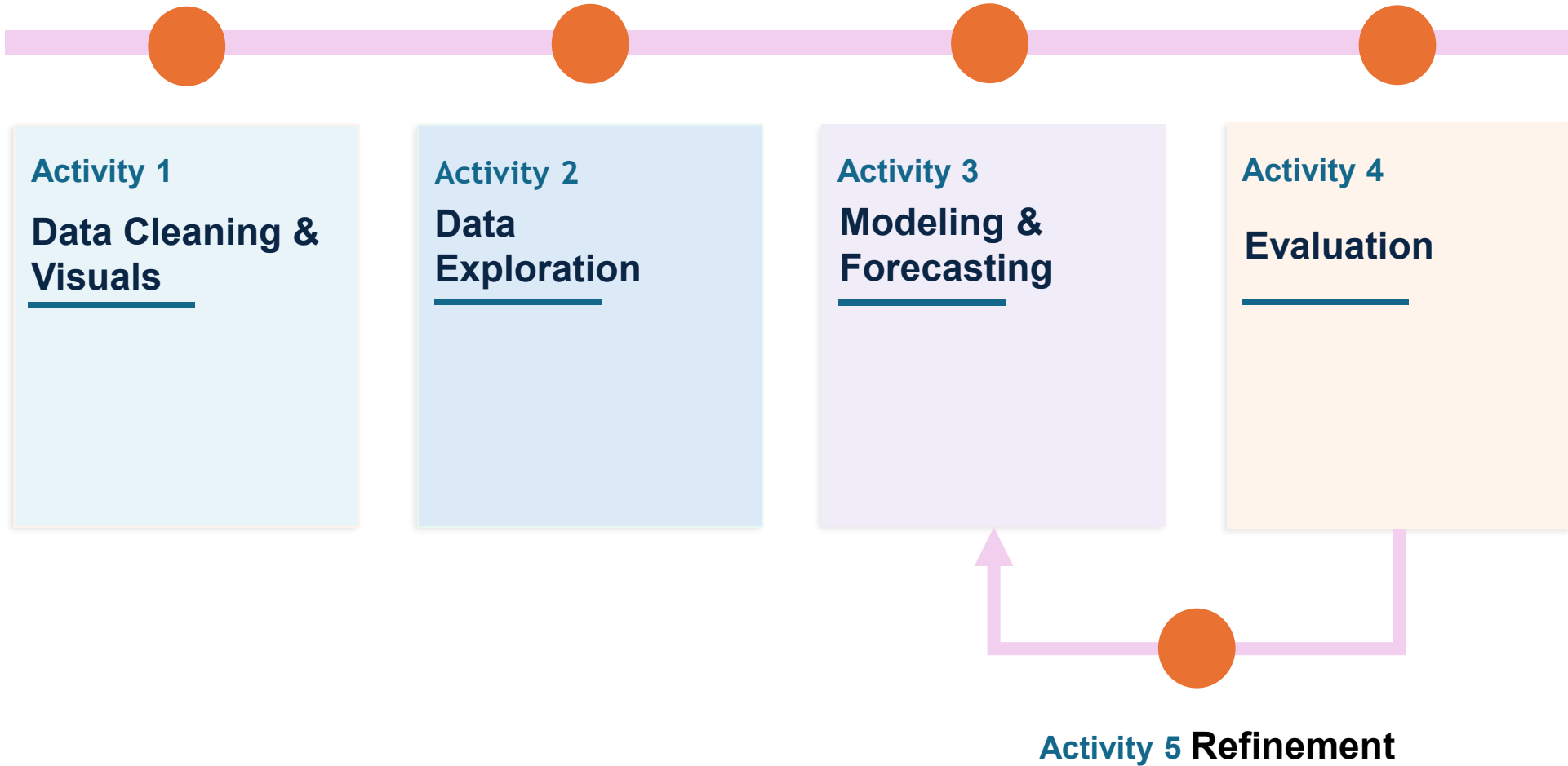
`agents.md`

Generated by AI

- AI-readable agent instructions
- Describes HOW to execute the pipeline
- The intermediary between you and code

At the end of the day, **you are still the expert** making the decisions.

Lab Timeline



General Lab Structure

The Human-AI Iteration Loop: For each lab phase, you will complete the following steps.

1. Edit the Rules

Open the rules doc and **write** or **review** the rules for the given phase.

2. Generate agents.md

Prompt Copilot to **build** an agent from your rules document.

3. Generate R Script

Use the agent to **generate** the **R script** for this phase.

4. Run & Evaluate

Execute the script in VS Code.
Check the output and results.

5. Update & Regenerate

Refine the rules → **regenerate** agents.md → **regenerate** script.

6. Push to GitHub

Push results to GitHub.

Lab Objectives + End Products

Objectives	End Products
Use AI as a collaborative tool to build, refine, and execute an infectious disease forecasting pipeline.	rules.md & agents.md files
Translate epidemiological knowledge into structured instructions that guide AI-generated code.	rules.md & agents.md files
Develop a model to forecast incident influenza hospitalizations at the national level using CDC NHSN data.	Forecasting model scripts & forecast outputs
Evaluate forecast performance and iteratively improve model quality through human-AI teaming.	rules.md & agents.md files; Forecasting model scripts & forecast outputs